

ADENIYI ADEYEMI SAMUEL

Senior Software Engr.

[Github](#)



[Portfolio](#)



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Professional Summary

Highly skilled Software Engineer with over seven years of hands-on experience building, scaling, and deploying production-grade web, mobile, and AI-powered systems. Specialized in backend engineering using Node.js, PHP and Python (FastAPI), cloud-native architectures on AWS, and real-time applications. Proven track record of leading teams, optimizing performance, and delivering mission-critical systems used by over **30,000 users**.

Technical Skills & Engineering Depth

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL, PHP

Backend Engineering: Node.js, Express.js, NestJS, FastAPI, Remix, Prisma, AWS CDK, Laravel; RESTful API design, microservices architecture, authentication/authorization (JWT, OAuth2), background jobs, rate limiting, and API security hardening.

Frontend Engineering: React.js, Next.js, React Native; responsive UI design, component-driven architecture, state management, accessibility, and performance optimization.

AI & Intelligent Systems: Gemini API integration, AI inference pipelines, prompt engineering, ML model consumption, real-time AI-assisted workflows, automation systems.

Databases & Storage: PostgreSQL, MySQL, MongoDB, SupabaseDB, Redis; schema design, indexing, query optimization, migrations, data integrity, caching strategies.

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, ECS, CloudFormation), Docker, CI/CD pipelines (GitHub Actions), monitoring, logging, backups, and disaster recovery.

Engineering Practices: Clean Architecture, Domain-Driven Design principles, Agile/Scrum, technical documentation, code reviews, test-driven development, AI-assisted development.

Certifications: AWS, Oracle, Docker and Linux Certified Cloud Engineers.

Professional Work Experience

EMGE Resources (Edge Healthcare ERP)

2026 [Link](#)

Full Stack Software Engineer/Platform Engineer

Technology Stack: TypeScript, NestJS, Next.js, Rust, Sequelize (ORM), PostgreSQL, Redis, Bun, Nx Mono Repo, MinIO/S3-Compatible Storage, Docker, Docker Compose, BullMQ, ABAC/Permissions, Tailwind CSS, Google/Discord Webhook

Summary: Engineered a hybrid cloud and on-premise multi-tenancy Enterprise Resource Planning (ERP) platform for hospitals and enterprise operations, designed to keep critical services running locally during internet outages while syncing selected data to the cloud for centralized oversight. The system supports identity, access control, HR, patient operations, EMR, finance, documents, reporting, notifications and administrative workflows from a single scalable codebase. It is a high level solution which supports internal infrastructure for deployment using GUI rust based developed installer because the onprem artifacts are varying OS compatible executable binaries.

Technical Highlights

Architecture & Infrastructure

- Built an offline-first ERP architecture (On-prem LAN Server not complete offline to avoid pseudo-ops) with one shared codebase supporting both cloud and hospital on-prem with capabilities separation boundary and deployments.

- Designed multi-tenancy scoped access control users, roles, permissions, memberships, organizations, and hospital nodes.
- Implemented modular backend services for identity, HR, patient workflows, EMR, registration, finance, reporting, documents, etc.
- Separated local hospital operations from cloud oversight to support real world low connectivity environments.
- Added deployment-aware configuration and runtime behaviour without forking the application.

Frontend UI & UX

- Implemented permission aware navigation and workflows that adapt to actors role, and essentially the given permissions/privileges in the system.
- Built best and most simplified UX for users even with high data but guided and easily navigable user interfaces without verbosity.

Data & Storage

- designed relational models for organizations, identities, memberships, staff, recruitment, EMR, documents, settings, finance, HR, and other operational records.
- Implemented append only audit flows across modules for sensitive actions.
- Added reusable object storage abstraction for on-prem which support filesystem or minIO and cloud compatible file handling. .

Key Metrics & Achievements

- Helped build a production/enterprise level solution capable of serving nations-world multi-hospitals, multi-role, low connectivity environments.
- Spear-headed system design, modularized architecture, features-by-features chunking and components by components workflows that moved the system from ideation to a final products.
- Improved reliability by ensuring hospital critical workflow can run locally without dependency on the cloud.
- Designed system with enterprise concerns in mind: offline resilience, modular scalability, auditability etc.

TeshTreats LLC

2026 [Link](#)

Full Stack Software Engineer/Platform Engineer

Technology Stack: TypeScript, Next.js, Prisma (ORM), PostgreSQL, Redis, OpenSearch, Docker, Docker Compose, BullMQ, Resend, Payment Gateway, Contabo VPS, Google/Discord Webhook

Summary: Built a fully functional commerce platform for TeshTreats LLC, supporting customer storefront shopping, admin operations, inventory management, order processing, manual payments, SEO indexing, and production deployment automation including daily DB backup.

Technical Highlights

Architecture & Infrastructure

- Designed a modular monorepo with separate web, API, shared types, workers Prisma Schema, and infrastructure scripts.
- Implemented permissions/privileges and role based access control for customers, admins, vendors, and super administrator with central login, temporary password provisioning, forced password changes, MFA workflows, etc.
- Deployed production API, worker, redis, opensearch, database, etc on a VPS using docker compose.
- Built server-side automation for deployment polling, database backups, rollback support, health checks, systemd timers, and slack/telegram deployment/backup reporting.

Data & Storage

- Modeled commerce data with Prisma and PostgreSQL, including users, roles, products, SKUs, inventory batches, orders, payments, notifications, sessions, password reset token, and other models.
- Added redis-backed background processing, and OpenSearch support for scalable search behaviour.

Key Metrics & Achievements

- Delivered production ready products for customers consumption.
- 12 to 24 hour database backup retention and cleanup policy. It also supports slack/telegram backup delivery and reporting.
- End to end product and inventory expiry alerts asides the system notification services.

Bug Bounty AI-Platform

2026

Solo Full Stack Engineer

Technology Stack: Go, Python, TypeScript and Next.js

Summary: Designed and implemented a **production-grade autonomous penetration testing platform** that automates the full bug-bounty workflow, from asset discovery to AI-powered vulnerability detection and professional security report generation. Built as a *distributed microservices architecture with 9 independently deployable services, event-driven communication, and a real-time dashboard*.

Technical Highlights

Architecture & Infrastructure

- Architected a **polyglot microservices system** with 5 Go services (high-performance networking, scanning, scheduling) and 4 Python services (AI reasoning, vulnerability classification, report generation).
- Designed ***event-driven inter-service communication*** using NATS JetStream for async task distribution and Redis for priority queue management.
- Implemented a fully automated end-to-end pipeline: Target Registration → Asset Discovery → Vulnerability Scanning → AI Classification → Exploit Generation → PDF/Markdown Report — with zero manual intervention between stages.
- Containerized all services with Docker and created Kubernetes Helm charts for production-grade orchestration

Backend Engineering (Go)

- API Gateway: Centralized reverse proxy with JWT authentication, RBAC middleware, dynamic service routing, and rate limiting
- Asset Discovery Engine: Parallel subdomain enumeration using DNS resolution (A/AAAA/CNAME/MX/NS/TXT), Certificate Transparency log parsing (crt.sh), Subfinder integration, and HTTP fingerprinting
- Scan Scheduler: *Redis-backed priority queue with NATS task distribution* to horizontally-scalable scanner workers; supports job cancellation, retry logic, and progress tracking
- Scanner Workers: Stateless, horizontally-scalable workers executing concurrent vulnerability tests (SQL Injection, XSS, Security Misconfigurations, Open Redirects, Authentication Bypass)

AI & Machine Learning (Python)

- Three-Stage Vulnerability Detection Pipeline: Pattern Matching (signature-based) → Heuristic Analysis (behavior-based) → LLM Reasoning (AI-augmented classification via configurable providers: OpenAI GPT-4, Ollama, vLLM)
- AI Exploit Generator: Generates proof-of-concept exploit payloads using LLM reasoning with safe-mode sandboxing, structured JSON extraction, and timeout handling for local model inference
- CVSS v3.1 Risk Scoring: Automated base score calculation with exploitability and impact sub-score breakdowns

Frontend & UX (Next.js + TypeScript)

- Built a *responsive real-time dashboard* with MobX state management, 8 switchable color themes, and a guided pipeline tracker that auto-navigates users through each stage of the pentest flow.
- Implemented toast notification system with *actionable navigation links*, skeleton loading states, and enhanced pagination
- Report downloads support PDF, Markdown, and JSON with per-report format regeneration

Data & Storage

- PostgreSQL 16 with optimized schema design, JSONB columns for flexible evidence storage, composite unique constraints for vulnerability deduplication, and efficient indexing strategies
- MinIO (S3-compatible) object storage for generated report files with authenticated download URLs

Key Metrics & Achievements

- **9 microservices** independently deployable and horizontally scalable.
- **5 scan modules** running in parallel across distributed worker cluster
- **3-stage AI pipeline** for vulnerability classification with configurable LLM backends
- **3 report formats** (PDF, Markdown, JSON) with professional segmented structure including executive summary, asset inventory, detailed findings with evidence, exploit PoCs, CVSS analysis, and remediation roadmap
- **Full end-to-end automation** single click triggers the entire pentest pipeline with real-time progress tracking

Waliin Social Networking Company

2025 – till date | Remote (US)

Senior Software Engineer (Backend-Focused)

Technology Stack: Nodejs, Next.js, Git, Javascript.

Core Responsibilities

Backend Development (Primary Focus)

- Design, develop, and maintain scalable RESTful APIs using Node.js for user-facing features
- Build and optimize the API layer handling user data, content delivery, and real-time interactions
- Implement secure authentication and authorization systems (JWT/OAuth)
- Develop database schemas and optimize queries for MongoDB/PostgreSQL/MySQL
- Create real-time features using WebSockets/[Socket.io](https://socket.io/) for notifications and messaging
- Implement caching strategies with Redis/Memcached for performance optimization
- Ensure API security, data protection, and compliance with privacy regulations

Admin Panel Development

- Build and maintain the administrative interface using PHP CodeIgniter framework
- Develop admin features for content moderation, user management, and analytics
- Create dashboard components for monitoring platform metrics and KPIs
- Implement role-based access control for different admin permission levels
- Develop reporting tools and data visualization for admin insights.

Lead Full Stack Developer

2023 - 2024

University of Africa, Toru-Orua.

Lead development of enterprise-grade Edutech App/Portal that is widely used by over 10,000 students and staff

Key Achievements

- Led development of enterprise academic platform serving **10,000+ students & staff**
- Designed microservices architecture handling **1M+ daily requests**
- Improved system performance by **70%** through caching and query optimization
- Implemented CI/CD pipelines reducing deployment time by **80%**
- Mentored and led a team of 4 developers.

Full Stack Engineer (Contract) - CGI Nederland

2022 – 2023

- Developed and documented REST APIs with strong data modeling practices
- Built responsive frontend interfaces using React and Tailwind CSS
- Collaborated with cross-functional teams in Agile environment

Medical Technology

2020 - 2022

Front End Engineer

- Built responsive, accessible healthcare interfaces using React and Tailwind
- Created reusable UI component systems with Shadcn/UI
- Improved cross-browser consistency and UX reliability

Software Engineer I

TOKU-E Company · Full-time

2019 - 2020

Hybrid

Project for predicting changes in the Forex Market. I was responsible for the whole project and had the roles from project manager to architect the application and assumed Lead developer role. The application was distributed and scalable multi-components system.

1. Connecting via REST API's to external systems.
2. Hiring and managing other developers.
3. Designing the architecture and setting up the backend user's layer (mobile app) using nodejs and admin layer in PHP.

Skills: *PHP · MySQL · Spring Boot · Node.js · React.js · User Experience (UX) · MERN Stack*

AI-Enabled Projects (Production Grade)

AI-Powered Emergency Care EMS

[Live Demo](#)

[Github](#)

Tech Stack: TypeScript, Supabase, Node/Express.js, React.js, GeminiAI API, SQL, and others

- Developed an intuitive UI with better user's experience.
- Integration of GeminiAI API to power emergency first aid assistance.
- Supabase Auth and backend logic to handle server side/api requests.
- Created standard emergency report generation modules.

NYSC Toru-Orua Automated Portal

[Live Demo](#)

[Github](#)

Tech Stack: Next.js, Gemini API, SupabaseDB, SQL.

- Design and develop a robust NYSC-Toru-Orua portal to save, post community development services activities.
- AI-powered platform that suggest community development program that specifically address the unique needs of people in Toru-Orua community.
- Automation of all NYSC Toru-Orua Members activities increasing work efficiency by 80%.
- Tracking of previous administration contributions leading to safe record keeping and others.

Personal Projects

Geriatrics Vital Signs Monitor (Mobile Application)

[Live Demo](#)

[Github](#)

Tech Stack: React Native, Node/Express.js, SQL, SupabaseDB.

- Developed custom simulation engine with configurable parameters
- Used React Native's native modules for optimal performance
- Implemented local storage with automatic sync capabilities
- Use NodeJs to handle API requests from the arduino kits to the mobile app

Portfolio

[Live Demo](#)

[Github](#)

AWS & DevOps Experience

- Designed and maintained cloud architecture on AWS, utilizing EC2, S3, RDS, Lambda, and CloudFormation.
- Implemented monitoring with CloudWatch and automated backups for disaster recovery.
- Set up CI/CD pipelines using GitHub Actions and AWS CodeDeploy for seamless deployments.
- Managed containerized applications with Docker and AWS ECS.

AI-Assisted Development

- Regularly use GitHub Copilot, Claude Code, Cursor, and Windsurf to accelerate development, reduce boilerplate code, and improve code quality.
- Implemented AI-assisted code review and documentation automation in team workflows.

Languages and Technologies

- **Languages:** JavaScript, TypeScript, Python, SQL, PHP
- **Frameworks & Tools:** React, Node.js, AWS, Azure Cloud, Docker, Git, Supabase, MongoDB, FastAPI, TensorFlow, Scikit-learn
- **Soft Skills:** Leadership, Project Management, Agile Development, Technical Documentation

Education, Practices & Professional Development

Bachelor of Science in Computer Science — Hanoi University, Vietnam (2024)

Relevant Coursework: Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Software Engineering.

Engineering Practices: Clean Architecture, Agile/Scrum, technical documentation, mentoring, AI-assisted development using GitHub Copilot, Cursor, Claude, and Windsurf.

Professional Strengths: System design, backend scalability, cross-team collaboration, technical leadership, and ownership of production systems.